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Recurrence Computational Fluid Dynamics (rCFD)

A Data-Driven Reduced-Order Acceleration Technique

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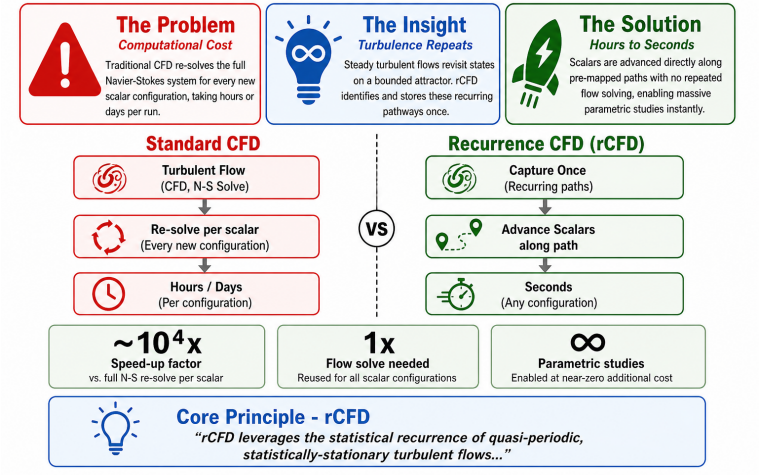
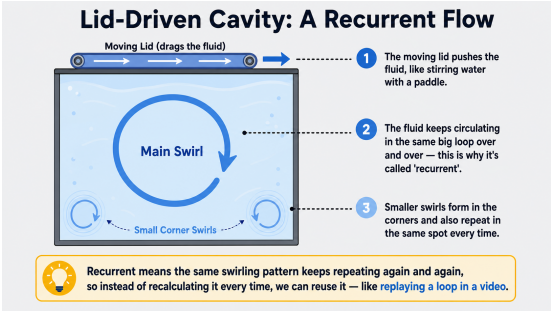
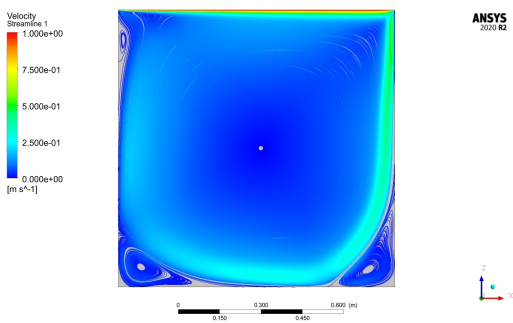
What is rCFD?

Motivation & Core Idea

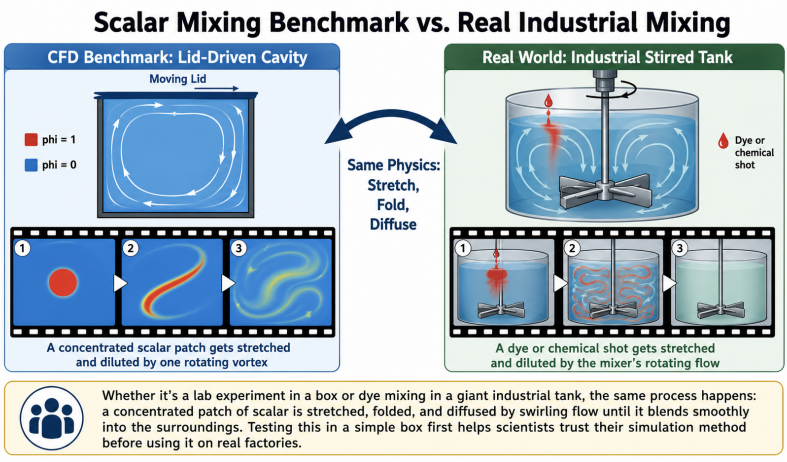


Why Lid Driven Cavity for Benchmarking?





The ANSYS simulation confirms the schematic: a dominant primary vortex fills the cavity core, with two smaller secondary vortices locked in the bottom corners. This matches the ”Main Swirl” and ”Small Corner Swirls” concept, proving the recirculation is a real, repeatable flow structure, not just an illustration. Once steady state is reached, this vortex pattern recurs at every time step — the exact recurrence rCFD exploits to skip repeated Navier-Stokes solves. This agreement validates the lid-driven cavity as the benchmark attractor for rCFD’s reduced-order reconstruction.

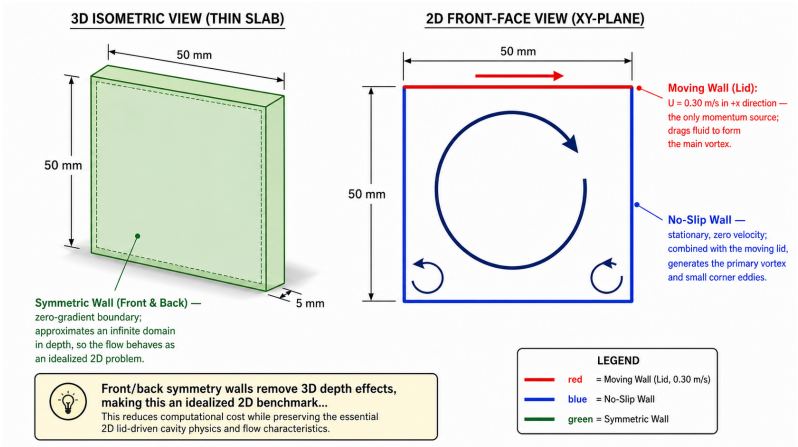


Pharma API Mixing: Impeller blends drug ingredients into excipients, stretching and diluting a dose for uniform potency.

Biotech Fermentation: Stirred bioreactors recirculate nutrients/reagents to keep cell cultures uniformly mixed.

Experimental Geometry, Boundary Conditions and Methods

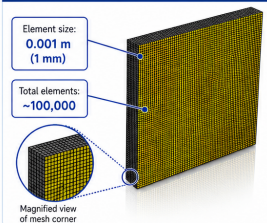
Geometry & Boundary Conditions



Fluid Property & Methods

NUMERICAL SETUP: DISCRETIZATION & SOLVER SETTINGS

MESH DISCRETIZATION



FLUID PROPERTIES: NITROGEN GAS AT 25°C

Parameter	Value
Density (ρ)	1.7 kg/m³
Dynamic Viscosity (μ)	1.76e-5 kg/(m·s)
Diffusivity (D)	2.5×10^{-5} m²/s
Reynolds Number (Re)	1000
Lid Velocity / Characteristic Velocity (U)	0.30 m/s
Characteristic Length / Cube Side (L)	0.05 m

SOLVER SETTINGS

TURBULENCE MODEL
k-omega SST

SOLUTION METHOD
SIMPLE

GRADIENT
Least Squares Cell-Based

PRESSURE
Second Order

MOMENTUM
Second Order Upwind

TRANSIENT FORMULATION
Second Order Implicit

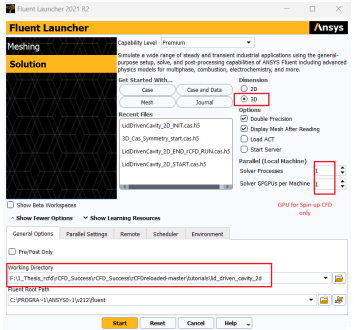
SUMMARY

- A fine, structured quadrilateral mesh with ~100,000 elements resolves boundary layers and gradients accurately on the thin slab.
- The k-omega SST model captures near-wall effects and adverse pressure gradients robustly.
- High-order discretization for pressure, momentum, and transient terms improves accuracy and reduces numerical diffusion.
- The SIMPLE algorithm with least-squares gradients ensures stable and reliable convergence.

Together, these choices yield accurate, stable, and efficient simulations of the flow over the thin square slab.

Steps of Running rCFD

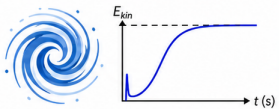
Start ANSYS Fluent and Load the .scm files



Start ANSYS Fluent and load the first `.scm` journal file; once it finishes running, close Fluent completely, relaunch it fresh, then load and run the next `.scm` file – repeating this restart cycle for each subsequent journal until all cases are processed.

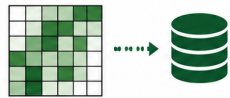
rCFD Workflow: Three Sequential Phases

1 Phase 1: Spin-Up CFD (CFD_start.scm)



- Solves unsteady Navier-Stokes (SIMPLE, k-omega, dt=0.01s) for 60s, monitors kinetic energy $E_{kin}(t)$ until it plateaus, outputs _START.cas.h5 baseline.

2 Phase 2: rCFD Database Prep (rCFD_prep.scm)

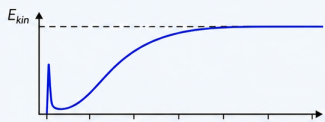


- Builds the recurrence database, computes cell shifts, local diffusivity, and a recurrence matrix.

3 Phase 3: Run rCFD (rCFD_run.scm)

- Reuses the stored recurrence database to rapidly reconstruct scalar transport and mixing.

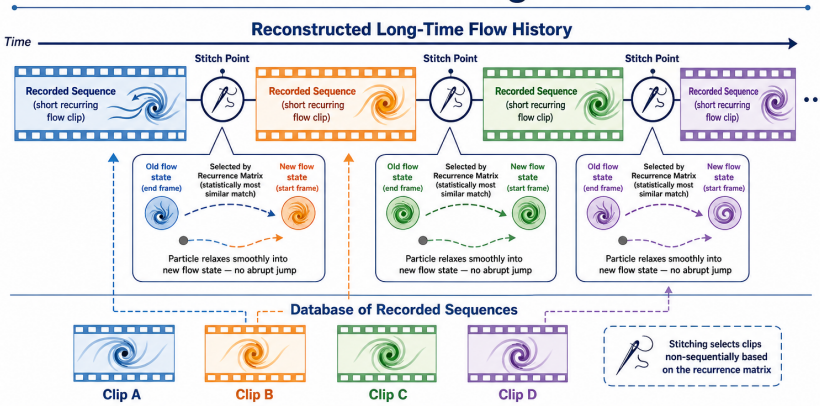
Why Spin-Up and Energy Stability Matter



Starting from a quiescent fluid creates **artificial transients**, corrupting the recurrence database.

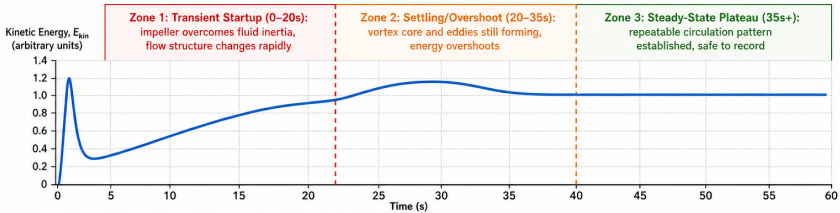
Monitoring kinetic energy confirms the flow reaches a **stationary attractor**. Reliable snapshots ensure **trustworthy rCFD results**.

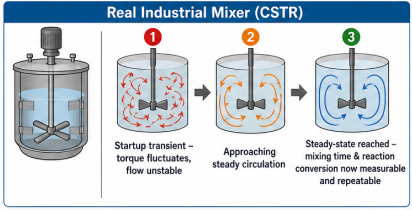
How rCFD Stitching Works

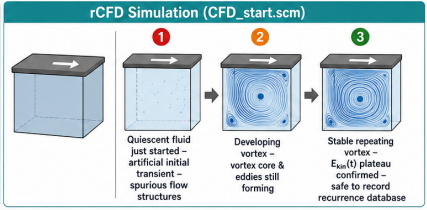


Instead of running one giant simulation, rCFD records **several short flow clips once**. To build a long-time history, it plays one clip, then **jumps ('stitches')** to the start of the **next best-matching clip** chosen from the **recurrence matrix**. Because the flow field changes abruptly at the jump, scalar-carrying particles are **not teleported** — they **relax smoothly** into the new flow state, **preserving physical accuracy**. This lets a few seconds of full CFD data represent **minutes or hours** of real process time at **fraction of the computational cost**.

Why Wait for Steady State? CSTR Startup Analogy





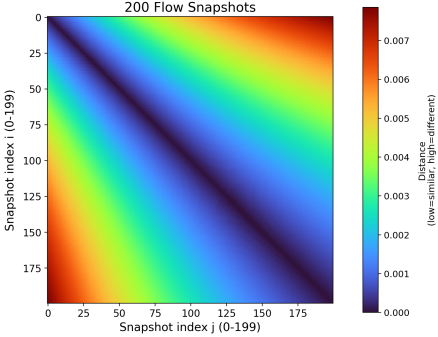


Consequence of Skipping This Step

Recording data too early –
non-repeating transient structures.
Wrong estimates in CSTR;
unreliable database in simulation.

Payoff of Waiting

Once steady-state is confirmed, the stable circulation pattern predicts mixing/conversion for many feed conditions in the reactor, or scalar transport for multiple injection scenarios using the rCFD database.



What It Shows

R(i,j) = dissimilarity between flow states i and j. Symmetric, with zero on the diagonal.

Dark Diagonal

Neighboring times look alike (smooth, continuous flow evolution).

Dark Off-Diagonal

Distant snapshots still match — the flow state repeats after a time lag.

Stripe Spacing

Reveals the flow's natural recurrence period.

Bright Zones

Opposite-phase states of the same repeating cycle — most dissimilar point in the loop.

Why This = Recurrence

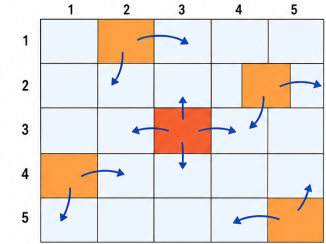
Regular banding (not random noise) proves the vortex genuinely revisits earlier states.

Why It Justifies rCFD

rCFD stitches statistically similar snapshots together instead of re-solving Navier-Stokes each time.

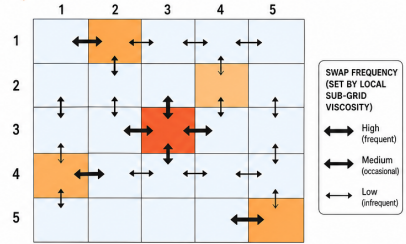
Inside rCFD: Convection vs Diffusion Mechanisms

CONVECTION: Cell-to-Cell Shift Vectors



Pre-computed shift vectors (extracted from stored velocity snapshots) tell each cell's scalar exactly where to move next along the stationary flow attractor — **replacing the need to solve the convection term** of Navier-Stokes equations directly.

DIFFUSION: Face-Swap Operations



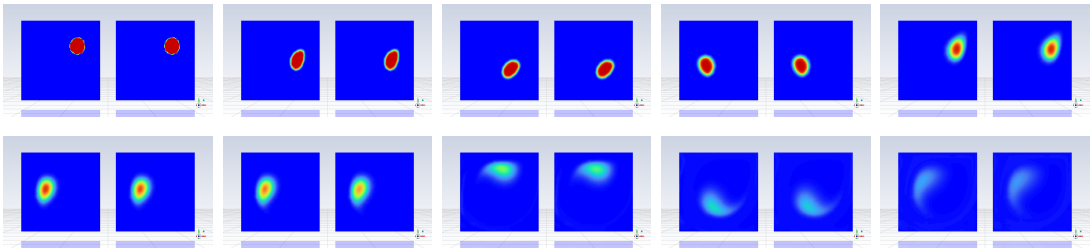
Instead of solving the diffusion equation, scalar concentration is **exchanged between neighboring cells** across shared faces. The frequency of these swaps is set by the local sub-grid viscosity, reproducing realistic turbulent mixing without extra computation.

Together, **cell-to-cell shifts (convection)** and **face-swap operations (diffusion)** let **rCFD** reproduce full scalar transport physics using only simple, precomputed bookkeeping operations on the mesh — **completely bypassing repeated Navier-Stokes solves.**

rCFD Prediction Results and Diffusion Progress Track

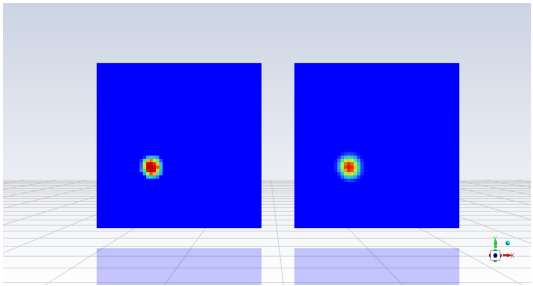
rCFD_user_run_topright.h (config header)	
Spec: Scalar transport, seeding (ϕ = 1)	**Target:** Homogenise toward ϕ = 0
Domain: 0.05 × 0.05 × 0.005 m	Centre: (C_x, C_y) = (0.035, 0.035) m
Radius: C_R = 0.005 m	Depth: $z \leq$ 0.005 m

Mixing progress tracked via rCFD contour snapshots

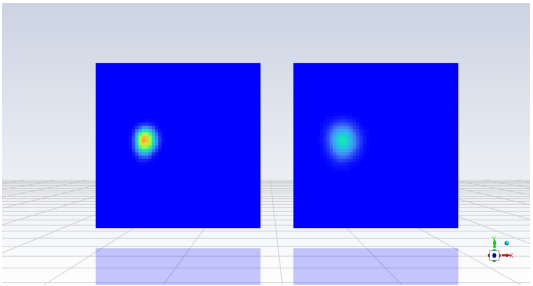


Infinite Predictions

Example Predictions using rCFD in Lid driven cavity



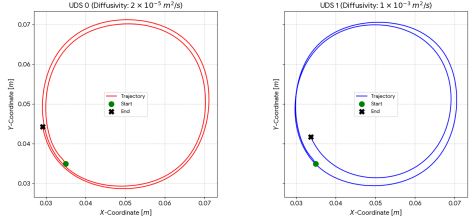
Initial State (Snapshot 0001)



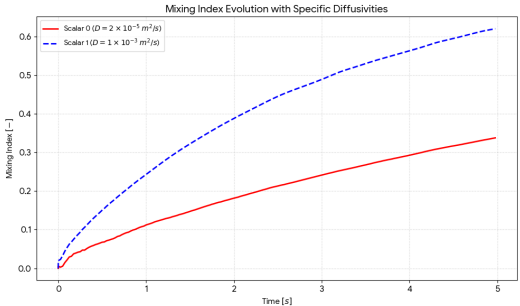
Later Mixing State (Snapshot 0013)

Visualization of scalar species mixing within the lid-driven cavity, transitioning from a localized initial condition to a more dispersed state.

Example Predictions using rCFD in Lid driven cavity



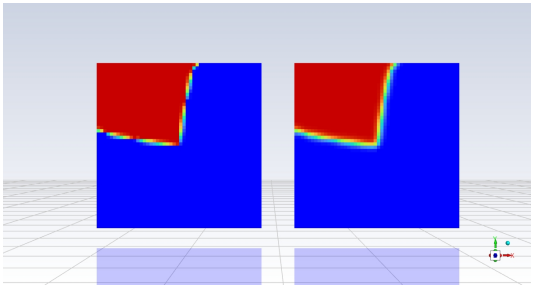
Center of Gravity (Cog) Evolution



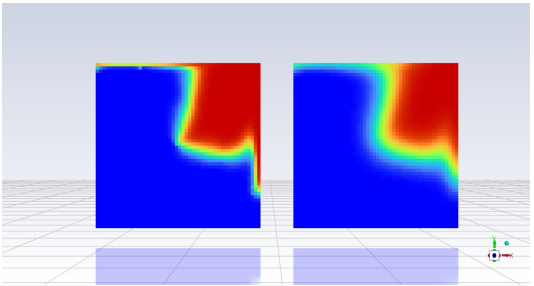
Mixing Index Evolution

Left: Tracking the global Center of Gravity of the species. Right: The mixing index approaching its asymptotic equilibrium value, demonstrating the ability of rCFD to capture long-term mixing dynamics.

Arbitrary Shape Placement & Fast Prediction



Initial Shape Definition (Snapshot 0001)

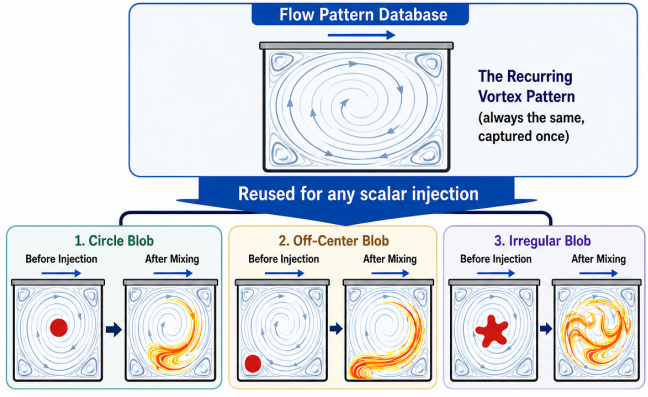


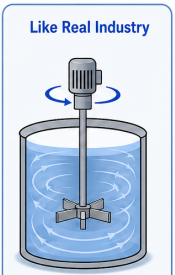
Predicted Diffusion State (Snapshot 0026)

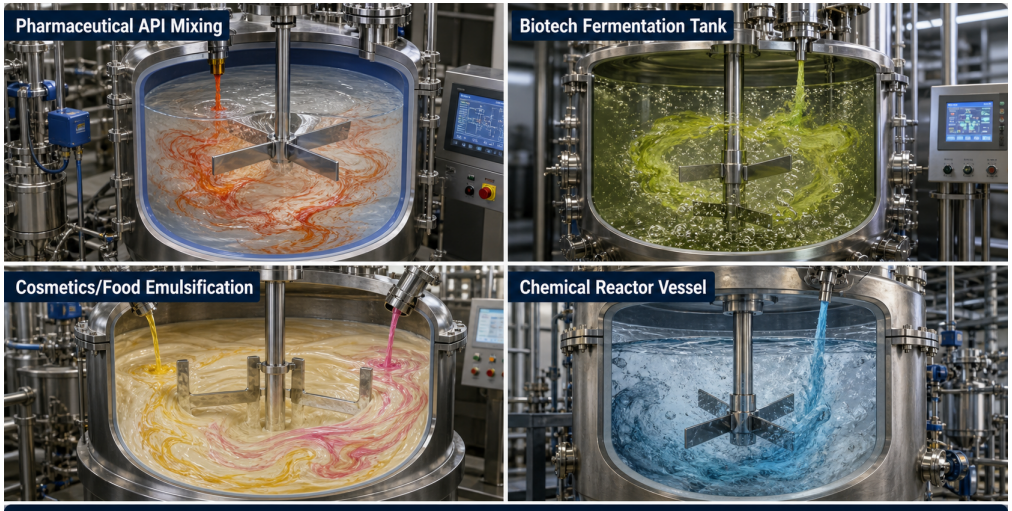
- **Any shape, any position:** Freely define arbitrary species concentration profiles anywhere in the flow domain without re-running the fluid solver.

Real-World Industrial Mixing Resembles the Lid-Driven Cavity

Flow Pattern Database






Pharmaceutical API Mixing
Biotech Fermentation Tank
Cosmetics/Food Emulsification
Chemical Reactor Vessel
All four processes share the same core physics as the lid-driven cavity benchmark:
a concentrated dose is stretched, folded, and diffused by recirculating flow until it becomes a uniform mixture.

Required Skills

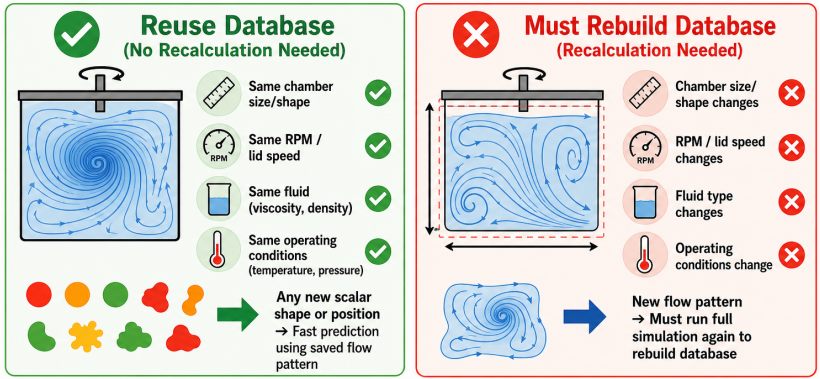
What You Need to Implement rCFD

- **Traditional CFD expertise:** Navier-Stokes solvers (SIMPLE), turbulence modelling (RANS, LES, DES).
- **Computational fluid programming:** UDFs, scripting (C, Scheme) for database automation.
- **Statistical analysis:** Convergence of leading statistical moments, recurrence prominence.
- **Validation capabilities:** Compare rCFD outputs against PIV experiments or detailed CFD predictions.

A multidisciplinary skill set – but the payoff is huge.

rCFD Database Validity: Fixed Setup = Infinite Reuse, Changed Setup = Recalculate

When Can We Reuse the Same rCFD Database?



The **rCFD flow pattern database** stays valid and **reusable** as long as the physical setup itself does not change: the **chamber's shape and size**, the **rotation speed (RPM)** or **lid speed**, the **fluid type**, and the **operating conditions** all stay fixed. **Only the scalar's shape, position, or type of dye/chemical** can change freely and still be predicted instantly using the same saved flow database. If **any physical parameter of the flow changes**, the vortex pattern is different, so the database must be **recalculated** from a new full simulation.

rCFD Limitations

When rCFD Fails (Bold Failures)

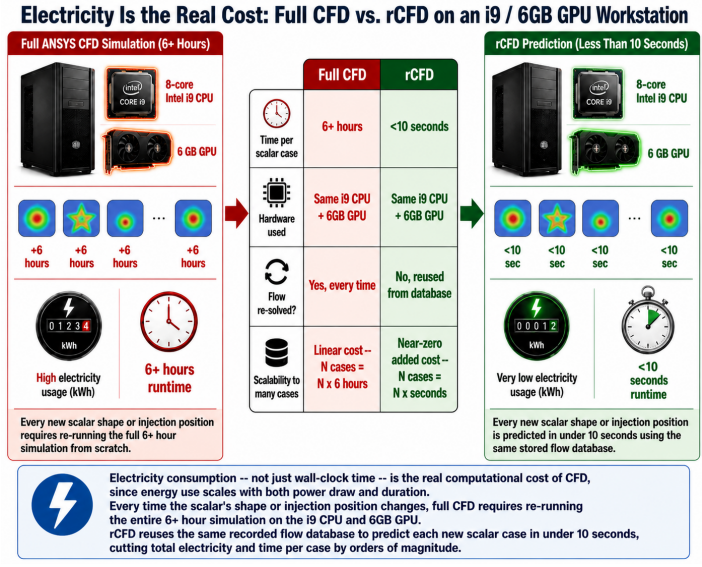
Cannot safely extrapolate

- Strong transients / start-up phases
- Chaotic turbulence without recurring patterns
- Changing geometry or fluid rheology

How to correct “bold failures”

- **Mass imbalance:** Mass-acting face swaps based on a global error monitor.
- **Numerical diffusion:** Account for local sub-grid viscosity and physical dispersion.
- **Low recurrence:** Domain decomposition into ”recurrence islands” + ”background sea”.

The True Computational Cost: Time, Power, and Scalability Compared


Electricity Is the Real Cost: Full CFD vs. rCFD on an i9 / 6GB GPU Workstation
Full ANSYS CFD Simulation (6+ Hours)
8-core Intel i9 CPU
6 GB GPU
+6 hours
+6 hours
+6 hours
+6 hours
01234
kWh
High electricity usage (kWh)
6+ hours runtime
Every new scalar shape or injection position requires re-running the full 6+ hour simulation from scratch.
Full CFD
rCFD
Time per scalar case
6+ hours
<10 seconds
Hardware used
Same i9 CPU + 6GB GPU
Same i9 CPU + 6GB GPU
Flow re-solved?
Yes, every time
No, reused from database
Scalability to many cases
Linear cost -- N cases = N x 6 hours
Near-zero added cost -- N cases = N x seconds
rCFD Prediction (Less Than 10 Seconds)
8-core Intel i9 CPU
6 GB GPU
<10 sec
<10 sec
<10 sec
<10 sec
00012
kWh
Very low electricity usage (kWh)
<10 seconds runtime
Every new scalar shape or injection position is predicted in under 10 seconds using the same stored flow database.
Electricity consumption -- not just wall-clock time -- is the real computational cost of CFD, since energy use scales with both power draw and duration.
Every time the scalar's shape or injection position changes, full CFD requires re-running the entire 6+ hour simulation on the i9 CPU and 6GB GPU.
rCFD reuses the same recorded flow database to predict each new scalar case in under 10 seconds, cutting total electricity and time per case by orders of magnitude.

Take-Home Messages

rCFD is a **game changer** for long-timescale transport simulations in recurrent flows.

Speed	Fidelity	Use case
100–10,000x faster	Statistical (not pointwise)	Digital twins, optimisation

- Works best when flow is statistically stationary and repetitive.
- Requires physics-based corrections (mass balance, diffusion, domain decomposition).

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